IGCSE 0607 Practice Questions — E5.7 — Combined Questions & Markscheme

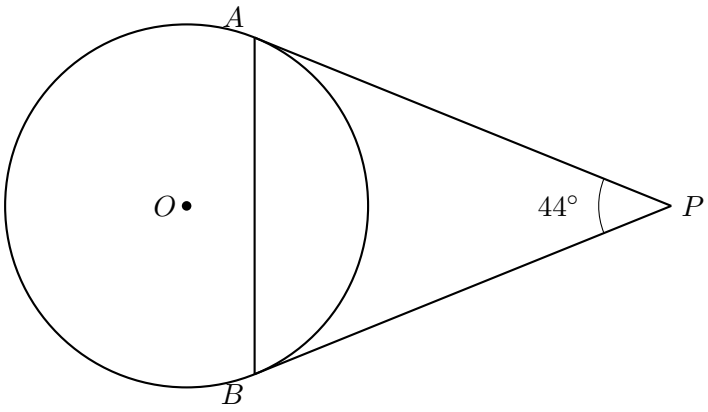
Circle theorems II

Total Questions: 15 — **Total Marks:** 95 Marks

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- **E5.7** Circle theorems II
  → **Paper 2 (Non-Calculator)** *(6 Qs)* · **Paper 4 (Calculator)** *(9 Qs)*

1 The diagram shows a circle with centre O.
P is an external point.
The lines PA and PB are tangents to the circle at the points A and B.
Angle $APB = 44°$.



NOT TO SCALE

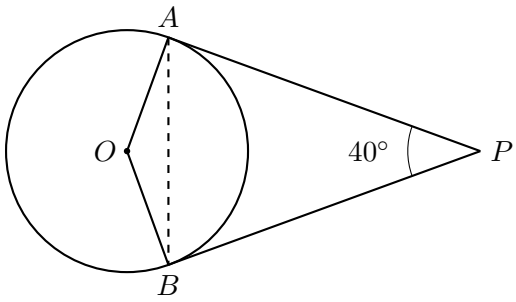
(a) Calculate angle PAB.

Angle PAB = ... ° **[2]**

(b) Calculate angle AOB.

Angle AOB = ... ° **[2]**

2 P is a point outside a circle, centre O.
PA and PB are tangents to the circle at A and B.
Angle $APB = 40°$.



(a) Write down the geometrical reason why triangle PAB is isosceles.

..... **[1]**

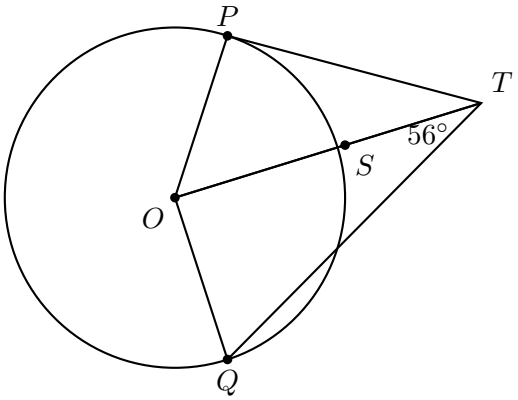
(b) Calculate angle PAB.

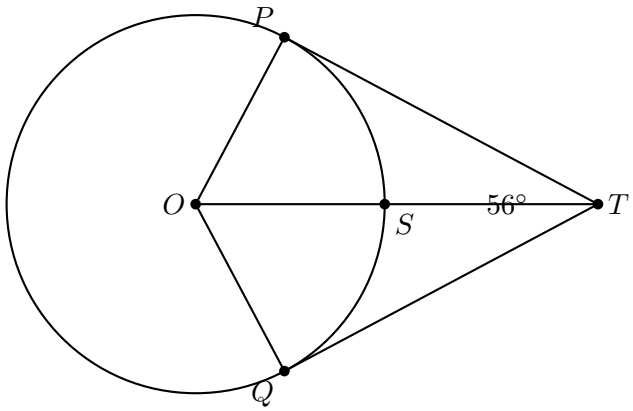
Angle PAB = ... ° **[1]**

(c) Calculate angle AOB.

Angle AOB = ... ° **[2]**

3 P and Q are points on a circle, centre O.
TP and TQ are tangents to the circle from an external point T.
The line TO intersects the circle at S.
Angle $PTQ = 56°$.





(a) Calculate angle POT.

Angle $POT =$ ..... ° **[2]**

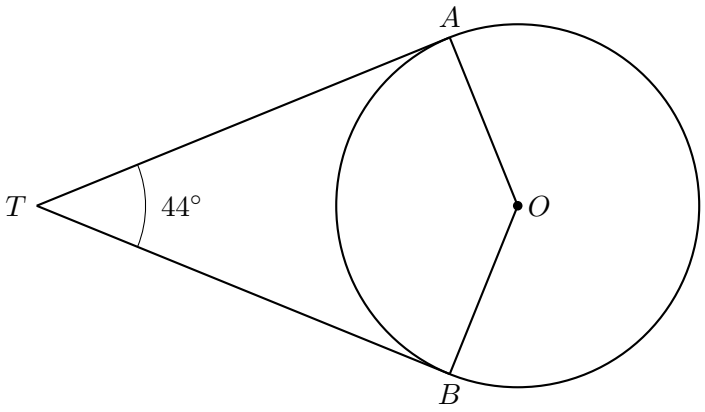
(b) Explain why triangle TPQ is isosceles.

..... **[1]**

(c) Calculate angle TPQ.

Angle $TPQ =$ ..... ° **[1]**

4 The diagram shows a circle, centre O.
TA and TB are tangents to the circle at A and B.
Angle $ATB = 44°$.



(a) Work out angle AOB.

Angle $AOB =$ ..... ° **[2]**

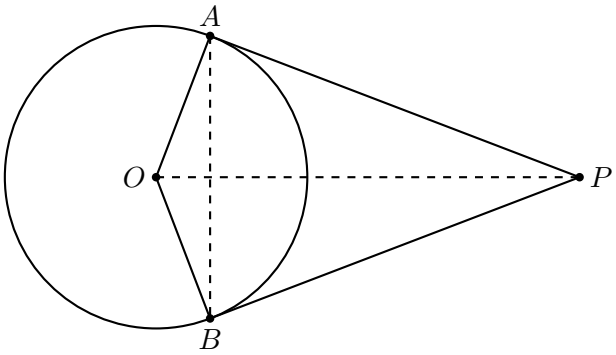
(b) Work out angle OAB.

Angle $OAB =$ ..... ° **[2]**

(c) Write down the geometrical reason why the length of TA is equal to the length of TB.

..... **[1]**

5 P is an external point.
PA and PB are tangents to a circle, centre O.
The line OP intersects the chord AB at right angles.
$OA = 7$ cm and $PA = 24$ cm.



(a) Write down the length of PB.

..... cm **[1]**

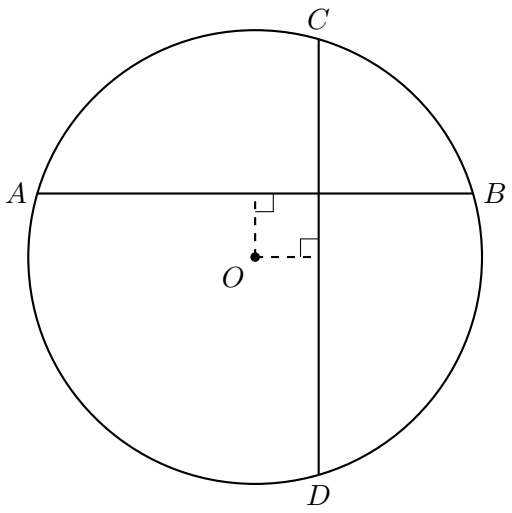
(b) Calculate the length of OP.

..... cm **[2]**

(c) Calculate the shortest distance from O to the chord AB.

..... cm **[3]**

6 The diagram shows a circle, centre O, with radius 25 cm.
AB and CD are two chords of the circle such that $AB = CD$.
The perpendicular distance from O to AB is $(x + 3)$ cm.
The perpendicular distance from O to CD is $(3x - 5)$ cm.



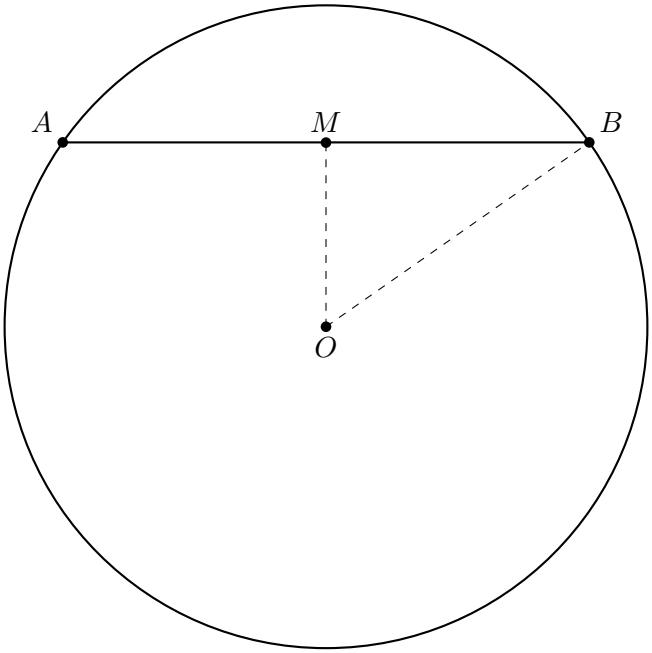
(a) Find the value of x.

$x =$ ..... **[2]**

(b) Calculate the length of the chord AB.

$AB =$ ..... cm **[3]**

7 A circle, centre O, has a radius of 8.5 cm.
A chord AB has length 14 cm.
M is the midpoint of AB.



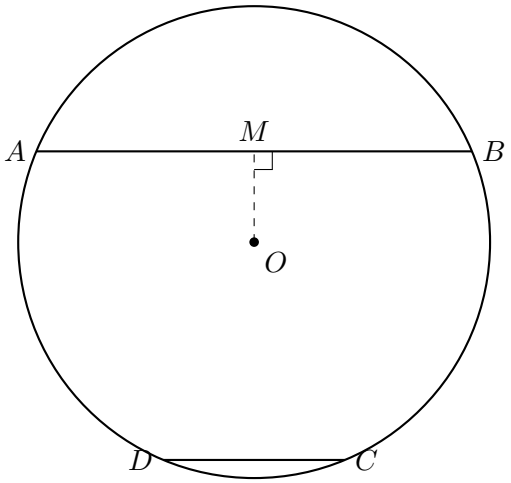
(a) Calculate the distance OM.

$OM =$ ..... cm **[3]**

(b) Another chord CD is drawn such that it is parallel to AB and is at a distance of 3 cm from O.
Calculate the length of the chord CD.

$CD =$ ..... cm **[2]**

8 The diagram shows a circle, centre O, with radius 13 cm.
AB is a chord of the circle of length 24 cm.
The point M is the midpoint of the chord AB.
Another chord CD is drawn inside the circle.



NOT TO SCALE

(a) Calculate the perpendicular distance, OM, from the centre of the circle to the chord AB.

$OM =$ ..... cm **[3]**

(b) The perpendicular distance from the centre O to the chord CD is also 5 cm.
Write down the length of the chord CD.

$CD =$ ..... cm **[1]**

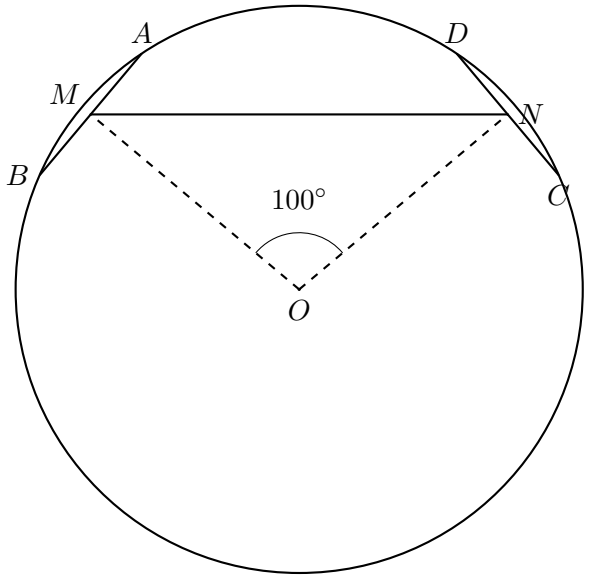
(c) Calculate the area of triangle OAB.

Area =

.....

cm^2 **[2]**

9 A, B, C and D are points on a circle, centre O and radius 25 cm.
The chords AB and CD each have a length of 14 cm.
M is the midpoint of AB and N is the midpoint of CD.
Angle $MON = 100°$.



(a) Show that $OM = 24$ cm.

[2]

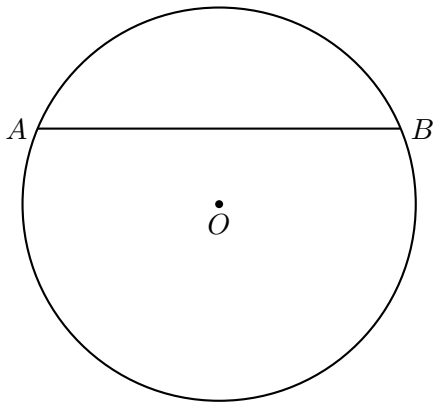
(b) Calculate the length of MN.

..... cm **[3]**

(c) Calculate the area of triangle MON.

..... cm^2 **[2]**

10 The diagram shows a circle, centre O, with radius 13 cm.
AB is a chord of the circle of length 24 cm.



(a) Calculate the shortest distance from O to the chord AB.

..... *cm* **[3]**

(b) CD is another chord of the circle.
CD is parallel to AB and has length 24 cm.
Write down the shortest distance from O to CD.

..... *cm* **[1]**

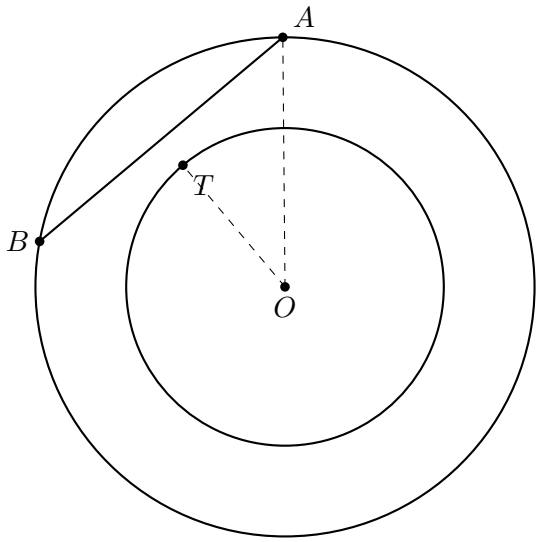
(c) EF is a third chord of the circle.
The shortest distance from O to EF is 12 cm.
Calculate the length of the chord EF.

..... *cm* **[3]**

(d) State the geometrical property of a circle used to find the length of half the chord in part (a).

..... **[1]**

11 Two circles, C_1 and C_2, have a common centre O.
The radius of C_1 is r cm and the radius of C_2 is R cm, where $R > r$.
A chord AB of the larger circle C_2 is tangent to the smaller circle C_1 at the point T.
The length of the chord AB is 24 cm and the area of the annular region between the two circles is $k\pi$ cm^2.



(a) Show that $R^2 - r^2 = 144$.

[3]

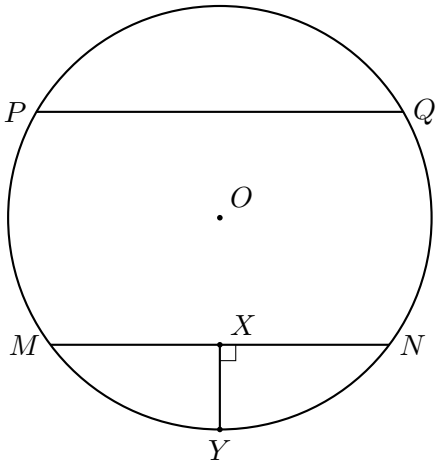
(b) Find the value of k.

$k =$ ..... **[2]**

(c) A second chord CD of the larger circle C_2 has length 24 cm.
State, with a reason, the distance from O to the chord CD.

..... **[2]**

12 A circle has centre O and radius 10 cm.
MN is a chord of length 16 cm.
The perpendicular bisector of MN intersects MN at X and the minor arc MN at Y.
PQ is a second chord which is parallel to MN.
The perpendicular distance between MN and PQ is 11 cm.



(a) Explain why the line containing XY passes through the centre O.

..... **[1]**

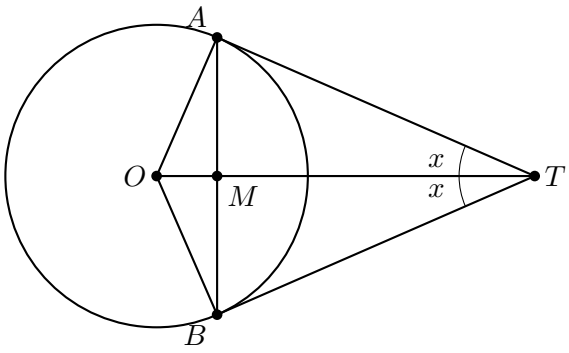
(b) Calculate the length of XY.

$XY =$ ..... cm **[3]**

(c) Calculate the length of the chord PQ.

$PQ =$ ..... cm **[4]**

13 TA and TB are tangents to a circle, centre O and radius r, at the points A and B.
The line TO intersects the chord AB at M.
Angle ATO = Angle $BTO = x$.



(a) Show that the length of the chord AB is $2r\cos x$.

[2]

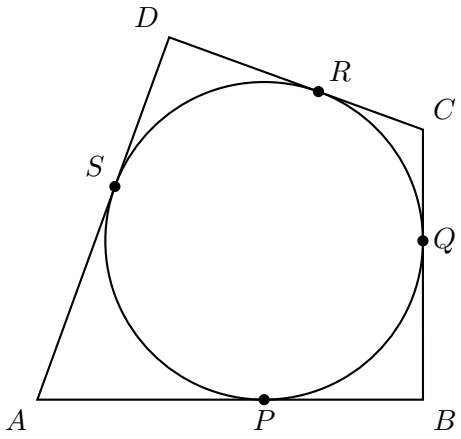
(b) A second chord BC is drawn such that $BC = AB$.
Write down the shortest distance from O to the chord BC, in terms of r and x.

..... **[1]**

(c) The area of triangle OAB is equal to the area of triangle ATB.
Find the exact value of x.

$x =$ ..... $^\circ$ **[5]**

14 A circle is inscribed in a quadrilateral $ABCD$.
The circle touches the sides AB, BC, CD, and DA at the points P, Q, R, and S respectively.



(a) Write down the geometrical reason why $AP = AS$.

..... **[1]**

(b) Show that $AB + CD = AD + BC$.

[3]

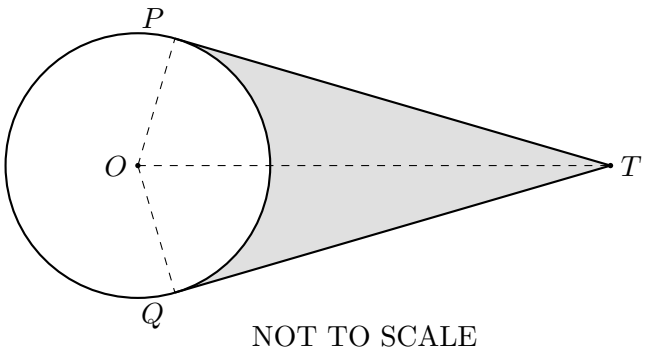
(c) Given that $AB = (2x + 5)$ cm, $BC = (3x - 2)$ cm, $CD = (x + 8)$ cm and $AD = (2x + 1)$ cm, work out the value of x.

$x =$ ..... **[3]**

(d) Calculate the perimeter of the quadrilateral $ABCD$.

Perimeter = ... cm **[2]**

15 The diagram shows a circle with centre O and radius 7 cm.
The lines TP and TQ are tangents to the circle at the points P and Q.
The distance from the centre O to the external point T is 25 cm.



NOT TO SCALE

(a) Show that the length of the tangent TP is 24 cm.

[2]

(b) Calculate angle PTQ.

Angle PTQ = ... ° **[3]**

(c) Calculate the area of the shaded region bounded by the tangents TP, TQ and the minor arc PQ.

Area =

.....

cm^2 **[4]**

1 **(a)** Recognises that tangents from an external point are equal in length, so triangle PAB is isosceles: $180 - 44 = 136$ **[M1]**

$136 \div 2 = 68$ **[A1]**

(b) Recognises angle between tangent and radius is $90°$, setting up sum of angles in quadrilateral $OAPB$: $360 - 90 - 90 - 44$ **[M1]**

136 **[A1]**

[Back to Question] *[Total Marks: 4]*

2 **(a)** Tangents from an external point are equal in length **[B1]**

(b) 70 **[B1]**

(c) Angle $OAP = 90°$ OR Angle $OBP = 90°$ **[M1]**

140 **[A1]**

[Back to Question] *[Total Marks: 4]*

3 **(a)** Angle $PTO = 56 \div 2 = 28°$ **[M1]**

$180 - 90 - 28 = 62$ **[A1]**

(b) Tangents from an external point are equal in length (so $TP = TQ$) **[B1]**

(c) 62 **[B1]**

[Back to Question] *[Total Marks: 4]*

4 **(a)** Uses angle sum of quadrilateral $OATB$ with angle $OAT = 90°$ and angle $OBT = 90°$: $360° - 90° - 90° - 44°$ OR $180° - 44°$ **[M1]**

136 **[A1]**

(b) Recognises triangle OAB is isosceles ($OA = OB$) and uses $(180° - 136°) \div 2$ **[M1]**

22 **[A1]**

(c) Tangents from an external point are equal in length **[B1]**

[Back to Question] *[Total Marks: 5]*

5 **(a)** 24 **[B1]**

(b) Identifies $\angle OAP = 90°$ (tangent perpendicular to radius) **[M1]**

$OP^2 = 7^2 + 24^2 = 625 \implies OP = 25$ **[A1]**

(c) Let M be the intersection of OP and AB. Identifies $\triangle OAM \sim \triangle OPA$ OR uses Area of $\triangle OAP$ OR uses $\cos(\angle AOP)$ **[M1]**

$\cos(\angle AOP) = \dfrac{7}{25}$ and $OM = 7 \times \cos(\angle AOP)$ **[M1]**

$OM = \dfrac{49}{25}$ or 1.96 **[A1]**

[Back to Question] *[Total Marks: 6]*

6 **(a)** $x + 3 = 3x - 5$ **[M1]**

$x = 4$ **[A1]**

(b) Perpendicular distance $= 7$ (can be implied) **[B1]**

$\sqrt{25^2 - 7^2}$ OR 24 **[M1]**

48 **[A1]**

[Back to Question] *[Total Marks: 5]*

7 **(a)** Angle $OMB = 90°$ (line from centre to midpoint of chord is perpendicular to chord) **[B1]**

$MB = 7$ **[B1]**

$OM = \sqrt{8.5^2 - 7^2}$ **[M1]**

4.77 or 4.769... **[A1]**

(b) Half chord $= \sqrt{8.5^2 - 3^2}$ **[M1]**

$2 \times \sqrt{63.25} = 15.9$ or 15.90... **[A1]**

[Back to Question] *[Total Marks: 5]*

8 **(a)** Identifies right-angled triangle OAM (or OBM) and states $AM = 24 \div 2 = 12$ **[B1]**

Applies Pythagoras' theorem: $OM^2 + 12^2 = 13^2$ **[M1]**

$OM = \sqrt{169 - 144} = \sqrt{25} = 5$ **[A1]**

(b) 24 (using property that chords equidistant from the centre are equal in length) **[B1]**

(c) $\frac{1}{2} \times 24 \times 5$ **[M1]**

60 **[A1]**

[Back to Question] *[Total Marks: 6]*

9 **(a)** States or uses $OM \perp AB$ (perpendicular bisector of chord passes through centre), so $AM = 7$ **[B1]**

Uses Pythagoras' Theorem: $OM = \sqrt{25^2 - 7^2} = \sqrt{625 - 49} = \sqrt{576} = 24$ **[B1]**

(b) States $ON = OM = 24$ (equal chords are equidistant from centre) **[B1]**

Uses Cosine rule: $MN^2 = 24^2 + 24^2 - 2(24)(24)\cos 100°$ OR uses $MN = 2 \times 24 \times \sin 50°$ **[M1]**

36.8 (accept $36.77\ldots$) **[A1]**

(c) $\frac{1}{2} \times 24 \times 24 \times \sin 100°$ **[M1]**

284 (accept $283.6\ldots$) **[A1]**

[Back to Question] *[Total Marks: 7]*

10 **(a)** Forms a right-angled triangle with hypotenuse 13 and base $\frac{24}{2} = 12$ **[M1]**

Uses Pythagoras' theorem: $\sqrt{13^2 - 12^2}$ **[M1]**

5 **[A1]**

(b) 5 **[B1]**

(c) Uses Pythagoras' theorem: $\sqrt{13^2 - 12^2}$ **[M1]**

Half of chord $EF = 5$ **[A1]**

10 **[A1]**

(d) The perpendicular from the centre to a chord bisects the chord (accept "perpendicular bisector of a chord passes through the centre") **[B1]**

[Back to Question] *[Total Marks: 8]*

11 **(a)** Angle $OTA = 90°$ (tangent perpendicular to radius) **[B1]**

$AT = 12$ (perpendicular from centre to chord bisects chord) **[B1]**

In triangle OTA, $r^2 + 12^2 = R^2$ (Pythagoras theorem) **[M1]**

$R^2 - r^2 = 144$ **[A1]**

(b) Area $= \pi R^2 - \pi r^2 = \pi(R^2 - r^2)$ **[M1]**

$k = 144$ **[A1]**

(c) Distance $= r$ (or same distance as chord AB) **[B1]**

Equal chords are equidistant from the centre **[B1]**

[Back to Question] *[Total Marks: 7]*

12 **(a)** The perpendicular bisector of a chord passes through the centre of the circle **[B1]**

(b) $MX = 16 \div 2 = 8$ **[B1]**

$OX = \sqrt{10^2 - 8^2} = 6$ **[M1]**

$10 - 6 = 4$ **[A1]**

(c) Distance of PQ from $O = 11 - 6 = 5$ **[B1]**

Half length of $PQ = \sqrt{10^2 - 5^2}$ **[M1]**

$\sqrt{75}$ or 8.66... **[A1]**

$2 \times \sqrt{75} = 17.3$ (17.32...) **[A1]**

[Back to Question] *[Total Marks: 8]*

13 **(a)** Identifies $\angle OAT = 90°$ and $\angle AOT = 90° - x$ **[M1]**

Uses $AM = r\sin(90° - x) = r\cos x$ and perpendicular bisector property $AB = 2AM$ to get $2r\cos x$ **[A1]**

(b) $r\sin x$ (equal chords are equidistant from centre, distance $= OM = r\cos(90° - x)$) **[B1]**

(c) Area of $OAB = \frac{1}{2} \times AB \times OM = \frac{1}{2}(2r\cos x)(r\sin x) = r^2 \sin x \cos x$ **[M1]**

$TM = \frac{AM}{\tan x} = \frac{r\cos x}{\tan x} = \frac{r\cos^2 x}{\sin x}$ **[M1]**

Area of $ATB = \frac{1}{2} \times AB \times TM = \frac{1}{2}(2r\cos x)\left(\frac{r\cos^2 x}{\sin x}\right) = \frac{r^2\cos^3 x}{\sin x}$ **[M1]**

Equates areas: $r^2 \sin x \cos x = \frac{r^2\cos^3 x}{\sin x} \implies \sin^2 x \cos x = \cos^3 x \implies \tan^2 x = 1$ **[M1]**

45 **[A1]**

[Back to Question] *[Total Marks: 8]*

14 (a) Tangents from an external point are equal in length **[B1]**

(b) Recognises $BP = BQ$, $CR = CQ$, and $DR = DS$ **[M1]**

$AB + CD = AP + PB + CR + RD$ **[M1]**

Substitutes equal tangent segments:
$AS + BQ + CQ + DS = (AS + DS) + (BQ + CQ) = AD + BC$ **[A1]**

(c) Substitutes expressions into the relationship from part (b):
$(2x + 5) + (x + 8) = (2x + 1) + (3x - 2)$ **[M1]**

Correctly expands and simplifies to a linear equation, e.g., $3x + 13 = 5x - 1$ **[M1]**

7 **[A1]**

(d) Substitutes $x = 7$ into the lengths of the sides (at least two) OR uses $2(3x + 13)$ **[M1]**

68 **[A1]**

[Back to Question] *[Total Marks: 9]*

15 (a) Recognises angle $OPT = 90°$ and sets up Pythagoras: $TP^2 + 7^2 = 25^2$ **[M1]**

$TP = \sqrt{625 - 49} = \sqrt{576} = 24$ **[A1]**

(b) Uses trigonometry in triangle OPT: $\sin(\text{angle } PTO) = \frac{7}{25}$ or $\tan(\text{angle } PTO) = \frac{7}{24}$ **[M1]**

Angle $PTO = 16.26°$ **[A1]**

Angle $PTQ = 2 \times 16.26° = 32.5°$ (accept $32.52°$) **[A1]**

(c) Calculates total area of kite $OPTQ = 2 \times (\frac{1}{2} \times 7 \times 24) = 168$ **[M1]**

Calculates angle $POQ = 360 - 90 - 90 - 32.52 = 147.48°$ (or uses $180 - 32.52$) **[M1]**

Calculates area of sector $OPQ = \frac{147.48}{360} \times \pi \times 7^2 = 63.07\ldots$ **[M1]**

Shaded area $= 168 - 63.07 = 104.93$ (accept 104.9 to 105.0) **[A1]**

[Back to Question] *[Total Marks: 9]*